

Eliminating Rework Through Departmental Quality Control

An upfitting line inspected its work once, at the end. Moving the check into the department that did the work removed one to two hours of assembly rework every day, and did it without adding a single inspector.

ORGANISATION	ROLE	SCOPE	PERIOD
Apex Manufacturing	Managing Director	Six stations, one line	2023, weeks 3 to 5

Where the rework came from

Quality was checked once, after paint, by people who had not built the unit. A wiring error introduced in the second hour of a four-day build was therefore found on day four, by which point three more stations had worked on top of it. Assembly was losing one to two hours every day to corrections of this kind, and the errors were largely upstream engineering ones rather than mistakes made at the bench being asked to fix them.

The cost was not only the hours. A defect found at the end is a defect nobody owns: the station that caused it has moved on, the station correcting it did not make it, and the line learns nothing either way. That is the part an extra inspector would not have fixed.

The change

- Each of the six stations - cut, weld, assembly, wiring, paint and final - was given written pass criteria for its own work, rather than one specification for the finished unit.
- Work stopped advancing until the department that did it signed it off. A unit that failed its own station's criteria went back to that station, not forward to the next one.
- The criteria were written by the departments that had to meet them. A standard handed down to a bench is a standard that bench works around.
- Supervisors were taught the reasoning behind each criterion rather than told which correction to apply, so a case the criteria did not anticipate still got judged correctly.
- End-of-line inspection was retired rather than kept alongside. Running both would have left the last check as the one that counted, and the station checks as paperwork.

What it changed

0 hrs

daily assembly rework,
from 1 to 2 hours

6 of 6

stations checking
their own work

0

inspectors added to
make the change

MEASURE	END-OF-LINE MODEL	DEPARTMENTAL MODEL
Where a defect is found	After paint	At the station that made it
Who corrects it	Whoever is at the end	Whoever made it
Daily assembly rework	1 to 2 hours	None
What the line learns	Nothing	The cause, at the bench
Inspection headcount	A final check	None separate

What it made possible

This was one of three changes made in the first fifty days, and on its own it did not restore the delivery record. What it did was remove the work that was not building anything. Once the line stopped rebuilding its own mistakes, the same output no longer needed the same number of hands, and the reduction from twenty-five people to thirteen followed from that rather than from a target. Every active account was delivering on schedule within fifty days.

Why it held

A quality standard usually leaves with the person who introduced it. This one did not belong to that person: it was written by the departments it governed, and the supervisors running it had been taught why each criterion existed rather than what to do when it failed. That is the difference between a rule a line follows and one it understands.

"He put the QC checks inside each department instead of at the end, which stopped us rebuilding other people's mistakes."

Tomas Duarte, Production Lead, Apex Manufacturing